

AI BlogNest API

AI-Powered Blog Managementand Content Generation REST API Platform

PHASE 1: BRAINSTORMING & IDEATION

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Submitted as a phase-wise project submission.

1. Problem Identification

Content creators and blog platform managers face notable operational friction due to fragmented publishing workflows and lack of intelligent automation. Key domain challenges include:

- **Fragmented Publishing & Content Management:** Inefficient mechanisms for organizing blog drafts, content categorization, tag taxonomies, and scheduled publications across decoupled environments.
- **Absence of Generative AI Assistance:** High effort and time consumption associated with drafting long-form content, summarizing extensive posts, and generating SEO-ready metadata.
- **Inadequate Security & Role Segregation:** Fragile access control models that lack clear boundaries between Administrators, Editors, Authors, and Readers.
- **Complex Moderation & Engagement Handling:** Cumbersome processes for moderating user discussions, approving articles, and monitoring user performance metrics.

2. Proposed Idea

AI BlogNest API is a robust, decoupled RESTful backend built with Node.js, Express.js, and MongoDB. It acts as an intelligent Content Management System (CMS) core that combines traditional structured blog publishing with modern Generative AI capabilities via Google Gemini AI.

3. Objectives

- **Generative Content Automation:** Integrate Google Gemini AI to dynamically synthesize high-quality, structured blog content from thematic prompts and produce rapid semantic summaries.
- **Stateless Security Infrastructure:** Implement stateless JSON Web Token (JWT) authentication and granular Role-Based Access Control (RBAC) to guard API entry points.
- **Scalable Architecture:** Architect a clean, maintainable Model-View-Controller (MVC) server layout separating data persistence, execution handling, and API route mapping.
- **Comprehensive CMS Management:** Provide full-featured CRUD routes for posts, categories, tags, user profiles, and moderated feedback comments.

4. Selected Features

- **AI Content Generation Engine:** Dedicated endpoint (`/api/ai/generate-blog`) accepting thematic inputs to output well-structured blog articles automatically.
- **Intelligent Content Summarizer:** Dedicated endpoint (`/api/ai/summarize`) parsing article payloads to return concise summaries.
- **Role-Based Route Shields:** Protected route channels configured specifically for Admin, Editor, Author, and Reader roles.
- **Data Validation & Security Middleware:** Parameter sanitization, request input validation, password hashing with `bcrypt.js` , and unified error handling.

5. Phase Outcome

A well-defined project scope, structural target, and functional blueprint establishing the baseline requirements and technical roadmap for the AI BlogNest API ecosystem.